VERTEX ALGEBRAS

github.com/danimalabares/vertex-algebras

Contents

1. Cartan subalgebra, Cartan matrix and Serre relations

Kac-Moody algebras are Lie algebras, whose definition is motivated by the structure of finite-dimensional simple Lie algebras over $\mathbb{C}$.

Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$. Then $\mathfrak{g}$ has a *Cartan subalgebra* $\mathfrak{h} \subset \mathfrak{g}$ (abelian + ...). Fixing $\mathfrak{h} \subset \mathfrak{g}$ gives a *root space decomposition*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$$

where $\Delta \subset \mathfrak{h}^*$ linear dual, and, by definition

$$\mathfrak{g}_\alpha = \{X \in \mathfrak{g} | [H, X] = \alpha(H)X \ \forall H \in \mathfrak{h}\}$$

Turns out the $\mathfrak{g}_\alpha$ are all 1-dimensional, though this property is lost when we go to Kac-Moody algebras.

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$$

The Killing form $\kappa : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$, $\kappa(x, y) = \mathrm{Tr}_{\mathfrak{g}} \mathrm{ad}(x)\mathrm{ad}(y)$ is nondegenerate. "This is kind of the definition of semisimple." (Think of $\mathfrak{h}$ as $\mathfrak{g}_0$, btw.)

$\kappa|_{\mathfrak{g}_\alpha \times \mathfrak{g}_\beta} \neq 0$ only when $\beta = -\alpha$. $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate. This gives a linear isomorphism $\mathfrak{h} \xrightarrow{\nu} \mathfrak{h}$ via $\nu(H)(H') = \kappa(H, H')$.

So, $\mathfrak{h}^*$ comes with a non-degenerate bilinear form.

The *reflection* $r_\alpha : \mathfrak{h} \to \mathfrak{h}^*$ in $\alpha \in \mathfrak{h}^*$ (usually a root) is $r_\alpha(\lambda) = \lambda - 2\frac{(\lambda,\alpha)}{(\alpha,\alpha)} \cdot \alpha$.

"Classify root systems [...] classify semisimple Lie algebras" It is a fact that $r_\alpha(\Delta) = \Delta$ for all $\alpha \in \Delta$, which motivates the definition of *root system* and permits classification.

Example 1.1. $\mathfrak{g} = \mathfrak{sl}_2$, $\mathfrak{h}$ = diagonal matrices

$$\begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \qquad \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix}$$

is a basis of $\mathfrak{h}$. There are 6 roots vectors

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

E_{23}, E_{13}, etc.

Exercise 1.2. $[H_1, E_{12}] = 2E_{12}$, $[H_2, E_{12}] = -E_{12}$, $\alpha_{12} = (2, -1)$.

[Drawing of roots]

Notions of *positive roots* and *simple roots* (set of rank$\mathfrak{g}$ simple roots has ℓ elements, where $\ell = \dim(\mathfrak{h}^*)$. This will also fail for Kac-Moody algebras more generally). Next write the *Cartan matrix*

$$A = (a_{ij}), \qquad a_{ij} = 2\frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}$$

for $1 \le i, j \le \ell$.

Example 1.3. $\mathfrak{sl}_3$. [Picture, hexagonal pattern]. $(\alpha_1, \alpha_1) = (\alpha_2, \alpha_2) = 2, (\alpha_1, \alpha_2) = -1$, so

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Example 1.4. $\mathfrak{sl}_5$. [Picture, square pattern]. $|\alpha_2| = 1$, $|\alpha_1| = 2$, $(\alpha_1, \alpha_2) = -2$, so

$$A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$

Remark 1.5. Since $\mathfrak{g}_\alpha$ is 1-dimensional, set $\mathfrak{g}_\alpha = \mathbb{C}E_\alpha$ and $E_i = E_{\alpha_i}$, $i = 1, 2, \ldots, \ell$ (simple root vectors). It turns out that

$$\mathrm{ad}(E_i)^{1-a_{ij}} E_j = 0.$$

This is called a *Serre relation*.

2. Some infinite dimensional Lie algebras

Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra, and define the *loop algebra*

$$\begin{aligned} L\mathfrak{g} &= \mathfrak{g}[t, t^{-1}], \text{ (with basis } at^m|^{a \,\in\, \text{a basis of } \mathfrak{g}}_{m \in \mathbb{Z}} \text{)} \\ &= \mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}] \end{aligned}$$

with the Lie bracket

$$[at^m, bt^n] = [a, b]t^{m+n}.$$

"This construction is absurdely general — we don't need $\mathfrak{g}$ to be semisimple [...]"

Take $\mathfrak{g} = \mathfrak{sl}_2$. Recall that

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

[Picture with $F, H, E, Ft, Ht, Et, Et^2 \ldots$] E was a root vector, corresponding to the unique root in $\mathfrak{sl}_2$, call it α_1. We seem to have a second simple root α_0, corresponding to Ft.

This looks like it wants to have a Cartan matrix

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

We will indeed recover (a variant of) $L\mathfrak{g}$ as a Lie algebra "built from" $A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, a Kac-Moody algebra. But note first, $\mathfrak{h} = \mathbb{C}H$ is too small. "Problem with α_0 and α_1 being linearly independent ..."

Exercise 2.1. Consider $L\mathfrak{g} \oplus \mathbb{C}d$, and set $[d, at^m] = mat^m$, $[d, d] = 0$. Check this defines a Lie algebra.

Proof. Skew-commutativity, i.e. for all $x \in L\mathfrak{g} \oplus \mathbb{C}d$,

$$[x, x] = 0, \tag{2.1.1}$$

is immediate from skew commutativity in $L\mathfrak{g}$ and the hypothesis that $[d, d] = 0$.

To confirm Jacobi identity, i.e. that for all $x, y, z \in L\mathfrak{g} \oplus \mathbb{C}d$

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0, \tag{2.1.2}$$

notice that since this is a cyclic sum on x, y, z we only need to consider three elements in $L\mathfrak{g} \oplus \mathbb{C}d$ up to cyclic permutation. The cases in which the three elements are either in $L\mathfrak{g}$ or in $\mathbb{C}d$ are obvious, so that there are only two interesting possibilities:

$$x = d, \qquad y = at^m, \qquad z = bt^n \tag{2.1.3}$$

$$x = d, \qquad y = d, \qquad z = at^n \tag{2.1.4}$$

Case 2.1.3 gives

$$\begin{aligned} &[d, [at^m, bt^n]] + [at^m, [bt^n, d]] + [bt^n, [d, at^m]] \\ &= [d, [a, b]t^{m+n}] + [at^m, -nbt^n] + [bt^n, mat^m] \\ &= (m+n)[a, b]t^{m+n} - n[a, b]t^{m+n} + m[b, a]t^{m+n} \\ &= (m+n)[a, b]t^{m+n} - n[a, b]t^{m+n} - m[a, b]t^{m+n} \\ &= (m+n)[a, b]t^{m+n} - (m+n)[a, b]t^{m+n} = 0. \end{aligned}$$

Case 2.1.4 gives

$$\begin{aligned} &[d, [d, at^m]] + [d, [at^m, d]] + [at^m, [d, d]] \\ &= [d, mat^m] + [d, -mat^m] = 0. \end{aligned}$$

□

The Kac-Moody algebra turns out to be, not quite this, but slightly larger still.

Recall that an *invariant bilinear form* $(\cdot, \cdot)$ on a Lie algebra $\mathfrak{g}$ is a bilinear form such that

$$([a, b], c) = (a, [b, c]) \qquad \forall a, b, c \in \mathfrak{g}. \tag{2.1.5}$$

Exercise 2.2. Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.

Proof. It's enough to show that $\mathfrak{g}$ is *perfect*, i.e. that $[\mathfrak{g},\mathfrak{g}] = \mathfrak{g}$. In this case, let $a, b \in \mathfrak{g}$ and suppose that $b = [x, y]$. Then

$$\begin{aligned}(a,b) &= (a,[x,y]) = (a,-[y,x]) = (-[a,y],x) = ([y,a],x)\\ &= (y,[a,x]) = (y,-[x,a]) = (-[y,x],a) = ([x,y],a) = (b,a)\end{aligned}$$

To confirm that $\mathfrak{g}$ is perfect just observe that $[\mathfrak{g},\mathfrak{g}]$ is a nontrivial ideal of $\mathfrak{g}$. □

Definition 2.3. Given $\mathfrak{g}$ simple, with $(\cdot,\cdot) : \mathfrak{g}\times\mathfrak{g} \to \mathbb{C}$ invariant bilinear form, the *affine Lie algebra* is

$$\hat{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K,$$

with $[K, \hat{\mathfrak{g}}] = 0$, and $[at^m, bt^n] = [a, b]t^{m+n} + m(a, b)\delta_{m,-n}K$.

"For the construction to work it doesn't actually have to be nondegenerate."

Exercise 2.4. Check that the affine Lie algebra $\hat{\mathfrak{g}}$ is a Lie algebra.

Proof. (Skew-commutativity.) Since $[K, \hat{\mathfrak{g}}] = 0$ and $K \in \hat{\mathfrak{g}}$, it is immediate that $[K, K] = 0$. For the case of an element in $L\mathfrak{g}$, we see that $[at^m, at^m] = 0$ by skew-commutativity of the bracket in $\mathfrak{g}$ and the Kronecker delta.

(Jacobi identity.) As in Exercise 2.1, any choice of x, y, z involving K is immediate by $[K, \hat{\mathfrak{g}}] = 0$. Thus the only interesting case is for Jacobi identity consider the cases

$$\begin{aligned}&[at^m,[bt^n,ct^\ell]] + [bt^n,[ct^\ell,at^m]] + [ct^\ell,[at^m,bt^n]]\\ &= [at^m,[b,c]t^{n+\ell} + n(b,c)\delta_{n,-\ell}K]\\ &+ [bt^n,[c,a]t^{\ell+m} + \ell(c,a)\delta_{\ell,-m}K]\\ &+ [ct^\ell,[a,b]t^{m+n} + m(a,b)\delta_{m,-n}K]\\ &= [at^m,[b,c]t^{n+\ell}] + [at^m, n(b,c)\delta_{n,-\ell}K]\\ &+ [bt^n,[c,a]t^{\ell+m}] + [bt^n,\ell(c,a)\delta_{\ell,-m}K]\\ &+ [ct^\ell,[a,b]t^{m+n}] + [ct^\ell, m(a,b)\delta_{m,-n}K]\\ &= [a,[b,c]]t^{m+(n+\ell)} + m(a,[b,c])\delta_{m,-(n+\ell)}K\\ &+ [b,[c,a]]t^{n+(\ell+m)} + n(b,[c,a])\delta_{n,-(\ell+m)}K\\ &+ [c,[a,b]]t^{\ell+(m+n)} + \ell(c,[a,b])\delta_{\ell,-(m+n)}K = 0\end{aligned}$$

It is clear that we obtain a Jacobi equation on $\mathfrak{g}$. To see that the remaining terms vanish, notice that the condition on the Kronecker delta in its three appearances is the same, namely, $m + n + \ell = 0$. In this case, we only need to check that $(a, [b, c]) = (b, [c, a]) = (c, [a, b])$ to conclude. This follows from the invariance of $(\cdot,\cdot)$ and the fact that $\mathfrak{g}$ simple using Exercise 2.2. □

We also have

Definition 2.5. The *extended affine Lie algebra* is

$$\tilde{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K \oplus \mathbb{C}d,$$

with $[d, at^m] = mat^m$ as before, and $[K, d] = 0$.

The extended affine Lie algebra is an example of a Kac-Moody algebra.

Exercise 2.6 (For those who like geometry)**.** Let $R = \mathbb{C}[t, t^{-1}]$. If $D \in \mathrm{Der}(R)$, then $L\mathfrak{g} \oplus \mathbb{C}d$ is a Lie algebra with $[d, a \otimes r] = a \otimes D(r)$. Is $(\mathfrak{g} \otimes R) \oplus \mathrm{Der}(R)$ a Lie algebra? (The Lie alegra $L\mathfrak{g} \oplus \mathbb{C}d$ from Exercise 2.1 is a particular case, for $D = t\frac{d}{dt}$.)

Proof. Checking that $L\mathfrak{g} \oplus \mathbb{C}d$ is a Lie algebra with $[d, a\otimes r] = a \otimes D(r)$ is similar to Exercise 2.1: skew-commutativity is immediate from skew-commutativity in each of the components, while Jacobi identity is verified in two cases. For $x = y = d$ and $z = a \otimes r$ we quickly obtain

$$\begin{aligned}
&[x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\
&= [d, [d, a \otimes r]] + [d, [a \otimes r, d]] + [a \otimes r, [d, d]] \\
&= [d, a \otimes D(r)] + [d, -a \otimes D(r)] = 0.
\end{aligned}$$

And for $x = d$, $y = a \otimes r$ and $z = b \otimes s$, we get

$$\begin{aligned}
&[x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\
&= [d, [a \otimes r, b \otimes s]] + [a \otimes r, [b \otimes s, d]] + [b \otimes s, [d, a \otimes r]] \\
&= [d, [a, b] \otimes rs] + [a \otimes r, -b \otimes D(s)] + [b \otimes s, a \otimes D(r)] \\
&= [a, b] \otimes D(rs) - [a, b] \otimes rD(s) + [b, a] \otimes sD(r) = 0.
\end{aligned} \tag{2.6.1}$$

To check whether $(\mathfrak{g} \otimes R) \oplus \mathrm{Der}(R)$ is a Lie algebra first put the Lie bracket on $\mathrm{Der}(R)$ as $[D, D_1] = DD_1 - D_1D$. It is clear that this bracket is skew-commutative. Jacobi identity reads

$$\begin{aligned}
&[D, [D_1, D_2]] + [D_1, [D_2, D]] + [D_2, [D, D_1]] \\
&= [D, D_1D_2 - D_2D_1] + [D_1, D_2D - DD_2] + [D_2, DD_1 - D_1D] \\
&= D(D_1D_2 - D_2D_1) - (D_1D_2 - D_2D_1)D + D_1(D_2D - DD_2) \\
&- (D_2D - DD_2)D_1 + D_2(DD_1 - D_1D) - (DD_1 - D_1D)D_2 \\
&= DD_1D_2 - DD_2D_1 - D_1D_2D + D_2D_1D + D_1D_2D - D_1DD_2 \\
&- D_2DD_1 + DD_2D_1 + D_2DD_1 - D_2D_1D - DD_1D_2 + D_1DD_2 = 0.
\end{aligned}$$

Now put the bracket on $(\mathfrak{g} \otimes R) \oplus \mathrm{Der}(R)$ as $[D, a \otimes r] = a \otimes D(r)$. Skew-commutativity is immediate. Jacobi identity for $x = D, y = a \otimes r$ and $z = b \otimes s$ is identical to the computation 2.6.1. In the case $x = D$, $y = D_1$ and $z = a \otimes r$, we get

$$\begin{aligned}
&[D, [D_1, a \otimes r]] + [D_1, [a \otimes r, D]] + [a \otimes r, [D, D_1]] \\
&= [D, a \otimes D_1(r)] + [D_1, -a \otimes D(r)] + [a \otimes r, [D, D_1]] \\
&= a \otimes DD_1(r) - a \otimes D_1D(r) - a \otimes [D, D_1](r) = 0
\end{aligned}$$

□

3. Kac-Moody algebras

Recall the notion of the free Lie algebra on a vector space V of generators (or a set X, think of V as a vector space with basis X):

Definition 3.1. The *free Lie algebra* on V is characterized by the universal property

$$\begin{tikzcd} V \arrow[rr, "f"] \arrow[dr, "i"'] & & \mathfrak{g} \\ & F(V) \arrow[ur, "\exists! \tilde{f}"'] & \end{tikzcd}$$

That is, for any linear map $f : V \to \mathfrak{g}$ with $\mathfrak{g}$ Lie algebra, there exists a unique $\tilde{f}$ homomorphism of Lie algebras $F(V) \to \mathfrak{g}$ such that $\tilde{f} \circ i = f$.

$$\mathrm{Hom}_{\mathrm{Lie}}(F(V), \mathfrak{g}) = \mathrm{Hom}_{\mathrm{Vec}}(V, \mathfrak{g})$$

naturally.

That is, F and the forgetful functor $G : \underline{\mathrm{Lie}} \to \underline{\mathrm{Vec}}$ are adjoint:

$$\mathrm{Hom}_{\underline{\mathrm{Lie}}}(F(V), \mathfrak{g}) \xrightarrow{\simeq} \mathrm{Hom}_{\underline{\mathrm{Vec}}}(V, G(\mathfrak{g}))$$

A realisation of $F(V)$. Let

$$T(V) = \mathbb{C} \oplus V \oplus V^{\otimes 2} \oplus V^{\otimes 3} \oplus \dots$$

be the tensor algebra of V.

Then inside $T(V)$ consider $F(V)$ the span of iterated commutators of elements of V.

Proposition 3.2. *This realises the free Lie algebra.*

Proof. In online notes. □

In the finite dimensional simple case, we had

$$a_{ij} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)},$$

which we think also as $\alpha_i, \alpha_j \in \mathfrak{h}^*$, and $\alpha_i^\vee = \frac{2}{(\alpha_i,\alpha_i)} \nu^{-1}(\alpha_i) \in \mathfrak{h}$.

Clearly, $\alpha_{ii} = 2$ for all i. a_{ij} might not equal a_{ji}, but certainly $a_{ij} = 0 \iff a_{ji} = 0$. And $\forall i \neq j$, $a_{ij} \leq 0$.

One further property. Set

$$\varepsilon_i = \frac{2}{(\alpha_i, \alpha_i)}, \quad \text{and} \quad D = \substack{\text{diagonal matrix} \\ \text{with entries } \varepsilon_i}$$

Then $A = DB$, where $B = ((\alpha_i, \alpha_j))$ is symmetric. If a matrix A is equal to (diag)(symm), we call it *symmetrizable.*

Definition 3.3. A *generalized Cartan matrix* is an integer matrix $A = (a_{ij})$ which is

- symmetrizable,
- $a_{ii} = 2$ for all i,
- $a_{ij} = 0 \iff a_{ji} = 0$,
- $a_{ij} \leq 0$ for $i \neq j$.

Definition 3.4. A *realisation* of a generalized Cartan matrix is a complex vector space $\mathfrak{h}$, and two sets

$$\Pi^\vee = \{\alpha_1^\vee, \alpha_2^\vee, \dots, \alpha_n^\vee\}, \quad \text{and,}$$
$$\Pi = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$$

such that $\langle \alpha_i^\vee, \alpha_j \rangle = a_{ij}$, $1 \le i, j \le n$.

Exercise 3.5. $\dim(\mathfrak{h}) \ge 2n - \operatorname{rank}(A)$.

Proof. For $A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, a realisation is given by

$$\Pi^\vee = \{H_1, H_0\}, \qquad \Pi = \{\alpha_0, \alpha_1\}$$

$$\mathfrak{h} = \mathbb{C}H, \mathbb{C}d, \mathbb{C}K,$$
$$\mathfrak{h}^* = \mathbb{C}\alpha_1 + \mathbb{C}\delta + \mathbb{C}\Lambda_0$$

(Canonical dual, $\langle \alpha_1, H \rangle = 2$, $\langle \delta, d \rangle = 1 = \langle \Lambda_0, K \rangle$, every other pairing 0.)

Then

$$\begin{cases} \alpha_1 = \alpha_1 \\ \alpha_0 = \delta - \alpha_1 \end{cases} \qquad \begin{cases} \alpha_1^\vee = H \\ \alpha_0^\vee = K - H \end{cases}$$

So we obtain

$$\langle \alpha_0^\vee, \alpha_1 \rangle = \langle K - H, \alpha_1 \rangle = 2$$
$$\langle \alpha_1^\vee, \alpha_0 \rangle = \langle H, \delta - \alpha_1 \rangle = -2$$
$$\langle \alpha_0^\vee, \alpha_0 \rangle = \langle K - H, \delta - \alpha_1 \rangle = +2$$

□

Finally let's define Kac-Moody algebras.

Let A be a generalized Cartan matrix. Let

$$\tilde{\mathfrak{n}}_+ = F(e_1, \dots, e_n),$$

the free Lie algebra on n generators, and similarly

$$\tilde{\mathfrak{n}}_- = F(f_1, \dots, f_n).$$

Let $\mathfrak{h}$ be a realisation of A. Set $\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}_- \oplus \mathfrak{h} \oplus \tilde{\mathfrak{n}}_+$.

Make $\tilde{\mathfrak{g}}(A)$ a Lie algebra by defining

- $[\mathfrak{h}, \mathfrak{h}] = 0$,
- $\forall H \in \mathfrak{h}$, $[H, e_i] = \langle \alpha_i, H \rangle e_i = \alpha_i(H) e_i$. And similarly, $[H, f_i] = -\alpha_i(H) f_i$.
- $[e_i, f_j] = \delta_{ij} \alpha_i^\vee$.

Then $\tilde{\mathfrak{g}}(A)$ is a Lie algebra (though not yet the Kac-Moody algebra). See Kac, [Kac90, Thorem 1.2].

Remark 3.6. In $\mathfrak{h}$ we have a lattice

$$Q^\vee = \mathbb{Z}\alpha_1^\vee + \ldots + \mathbb{Z}\alpha_n^\vee, \quad \text{and}$$
$$Q = \mathbb{Z}\alpha_1 + \ldots + \mathbb{Z}\alpha_n \text{ in } \mathfrak{h}^*$$

(root and coroot lattices). $\tilde{\mathfrak{g}}(A)$ is naturally Q-graded, with

$$\tilde{\mathfrak{g}}(A)_\beta = \operatorname{span}\{\text{commutators of } e_i \text{ with } \sum \alpha_i = \beta\}.$$

$\tilde{g}(A) = \mathfrak{h}$.

Theorem 3.7 (Gabber-Kac). *Denote by $I \subset \tilde{\mathfrak{g}}(A)$ the maximal Q-graded ideal, such that $I \cap \mathfrak{h} = \{0\}$. Then I is generated by the Serre relations*

$$ad(e_i)^{1-a_{ij}} e_j \qquad and \qquad ad(f_i)^{1-a_{ij}} f_j, \ i \neq j.$$

Proof. [Kac90, Theorem 9.11]. □

(The existence of the ideal I does not need the theorem; the importance of the theorem is providing an expression for the generators.)

Definition 3.8. The *Kac-Moody algebra* $\mathfrak{g}(A)$ is $\tilde{\mathfrak{g}}(A)/I$.

4. Affine Kac-Moody algebras

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, with $(\cdot,\cdot) : \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ invariant bilinear form,

$$([x,y],z) = (z,[y,z]) \qquad \forall x,y,z \in \mathfrak{g}$$

(Eg. the Killing form $\kappa(x,y) = \mathrm{Tr}_{\mathfrak{g}} \mathrm{ad}(x)\mathrm{ad}(y)$ is invariant.)

Typically we normalise $(\cdot,\cdot)$ so that $(\alpha,\alpha) = 2$ for the long roots of $\mathfrak{g}$.

Then $\hat{\mathfrak{g}} = \mathfrak{g}[t,t^{-1}] \oplus \mathbb{C}K$ (affine Lie algebra),

$$[at^m, bt^n] = [a,b]t^{m+n} + m\delta_{m,-n}(a,b)K, \qquad [K,\hat{\mathfrak{g}}] = 0$$

and $\tilde{\mathfrak{g}} = \hat{\mathfrak{g}} \oplus \mathbb{C}d$, $[d,K] = 0$, $[d,at^m] = mat^m$, (affine Kac-Moody algebra or "extended affine Lie algebra")

Theorem 4.1. *$\tilde{\mathfrak{g}}$ is a Kac-Moody algebra.*

Let $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha\in\Delta} \mathfrak{g}_\alpha$, ($\mathfrak{g}_\alpha = \mathbb{C}E_\alpha$.)

The simple roots and coroots. $\tilde{\mathfrak{h}} = \mathfrak{h} \oplus \mathbb{C}K \oplus \mathbb{C}d$. We identify $\tilde{\mathfrak{h}}^*$ with $\mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta$ where

$$\begin{aligned} \Lambda_0(\mathfrak{h}) &= \delta(\mathfrak{h}) = 0 \\ \Lambda_0(d) &= \delta(K) = 0 \\ \Lambda_0(K) &= \delta(d) = 1 \end{aligned}$$

The *real coroots* are

$$\hat{\Delta}^{V,re} = \{E_\alpha t^m | \alpha \in \Delta, m \in \mathbb{Z}\}$$

and there are also imaginary roots and coroots

$$\hat{\Delta}^{V,im} = \{Ht^m | H \in \mathfrak{h}, m \in \mathbb{Z} \setminus \{0\}\}$$

Roots:

$$\begin{aligned} \hat{\Delta}^{re} &= \{\alpha + m\delta | \alpha \in \Delta, m \in \mathbb{Z}\} \\ \hat{\Delta}^{im} &= \{m\delta | m \neq 0\} \end{aligned}$$

Xt^m:

$$\begin{aligned} [H, Xt^m] &= [H,x]t^m, \qquad H \in \mathfrak{h} \\ [K, xt^m] &= 0 \\ [d, xt^m] &= mxt^m \end{aligned}$$

so it $x \in \mathfrak{g}_\alpha$, $xt^m \in \tilde{\mathfrak{g}}_{\alpha+m\delta}$.

The invariant bilinear form $(\cdot,\cdot)$ from $\mathfrak{g} \times \mathfrak{g}$ extends uniquely to $(\cdot,\cdot) : \tilde{\mathfrak{g}} \times \tilde{\mathfrak{g}} \to \mathbb{C}$.

$(d,d) = (K,K) = 0$, $(d,K) = 1$ and $(d,\mathfrak{h}) = (K,\mathfrak{h}) = 0$.

So, in $\tilde{\mathfrak{h}}^*$:

$$\begin{aligned}(\Lambda_0, \Lambda_0) &= (\delta, \delta) = 0\\ (\Lambda_0, \mathfrak{h}^*) &= (\delta, \mathfrak{h}^*) = 0\\ (\Lambda_0, \delta) &= 1.\end{aligned}$$

Hence, $|\alpha + m\delta|^2 = |\alpha|^2$, $|m\delta|^2 = 0$.

Example 4.2. $\widetilde{\mathfrak{sl}_2}, \tilde{\mathfrak{h}}^* = \operatorname{span}\{\alpha, \Lambda_0, \delta\}$ with Gram matrix ...

We can make a choice of positive roots,

$$\hat{\Delta}_+ = \{\alpha + m\delta | \alpha \in \Delta, m > 0\} \cup \{m\delta | m > 0\} \cup \Delta_+$$

Obviously, if $\alpha \in \Delta_+$ is simple, $\alpha \in \hat{\Delta}_+$ is simple.

Notation. Let $\theta \in \Delta_+$ be a the highest root. ($\nexists \alpha \in \Delta_+$ such that $\alpha - \theta \in \mathbb{Z}_+\Delta_+$.) and $\alpha = \delta - \theta$.

Then $\alpha_0 \in \hat{\Delta}_+$ is simple and the set of simple roots is $\hat{\Pi} = \{\alpha_0, \underbrace{\alpha_1, \ldots, \alpha_\ell}_{\text{the finite simple roots}}\}$.

where $\ell = \operatorname{rank}(\mathfrak{g})$.

The coroot corresponding to α_0 is

$$\alpha_0^\vee = K - \theta^\vee, \qquad \theta^\vee = \frac{2}{(\theta, \theta)}\nu^{-1}(\theta) \in \mathfrak{h}$$
$$\text{and} \qquad E_{\alpha_0} = E_{-\theta}t.$$

5. Weyl group

Upshot. The Weyl group is a semidirect product of pseudoreflections and translations.

In any Kac-Moody algebra, we have

$$\begin{aligned}\text{roots} \qquad \Pi &= \{\alpha_1, \ldots, \alpha_\ell\} \subset \mathfrak{h}^*\\ \text{coroots} \qquad \Pi^\vee &= \{\alpha_1^\vee, \ldots, \alpha_\ell^\vee\} \subset \mathfrak{h},\end{aligned}$$

and *reflections* $r_i \in \mathrm{GL}(\mathfrak{h}^*)$, defined by

$$r_i(\lambda) = \lambda - \langle \lambda, \alpha_i^\vee \rangle \, \alpha_i.$$

One can check that

$$(r_i\lambda, r_i\mu) = (\lambda, \mu) \qquad \forall \lambda, \mu \in \mathfrak{h}^*$$

The *Weyl group* W is $\langle r_i | i = 1, \ldots, \ell \rangle \subset \mathrm{GL}(\mathfrak{h}^*)$.

Example 5.1. For $\widetilde{\mathfrak{sl}_2}$, r_1 is easy,

$$\begin{aligned}r_1(\alpha) &= -\alpha \qquad (\text{as in } \mathfrak{sl}_2)\\ r_1(\delta) &= \delta, \qquad r_1(\Lambda_0) = \Lambda_0.\end{aligned}$$

To compute r_0 take an arbitrary element $m\alpha_1 + k\Lambda_0 + f\delta$ and do:

$$\begin{aligned}r_0(m\alpha_1 + k\Lambda_0 + f\delta) &= m\alpha_1 + k\Lambda_0 + f\delta - \langle \alpha_0^\vee, m\alpha_1 + k\Lambda_0 + f\delta \rangle \, \alpha_0\\ \alpha_0 = \delta - \alpha_1, &\qquad \alpha_0^\vee = K - \alpha^\vee\end{aligned}$$

so we obtain

$$\begin{aligned} &= m\alpha_1 + k\Lambda_0 + f\delta - (k-2m)(\delta - \alpha_1) \\ &(k-m)\alpha_1 + k\Lambda_0 + (f-k+2m)\delta. \end{aligned}$$

Relative to basis $\{\alpha_1, \Lambda_0, \delta\}$.

$$r_1 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, r_0 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} m \\ k \\ f \end{pmatrix}$$

$$t = r_1 r_0 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}$$

Notice that δ is fixed by all r_i. Also $m\alpha + k\Lambda_0 + f\delta$, the *coefficient* of Λ_0 is fixed by all r_i.

Then

$$t(m\alpha_1 + k\Lambda_0 + f\delta) = (m-k)\alpha_1 + k\Lambda_0 + (f-k+2m)\delta.$$

Think of t as a translation.

The number k in

$$\mathfrak{h}^* \ni \hat{\lambda} = \lambda + k\Lambda_0 + f\delta$$

is called the *level* of $\hat{\lambda}$.

$\hat{\mathfrak{h}}$ = union of (hyper)planes of constant level which are stable under W. The roots α are all of level 0.

[Picture] "r_1 changes the sign of the finite path". And $t = r_1 r_0$ is a sort of translation. Indeed, in general we can consider $t_{\alpha_i} = r_{\alpha_i} \circ r_0 \in W$,

$$t_\alpha(\beta + m\delta) = \beta + (m + (\beta, \alpha_i))\delta$$

One can describe the action of t_α on $\hat{\lambda}$ in general (e.g. see [Kac90, Chapter 6])

Proposition 5.2. *For the affine Kac-Moody algebra $\hat{\mathfrak{g}}$ with $\hat{W} = \langle r_0, r_1, \ldots, r_\ell \rangle$ its Weyl group (and $W = \langle r_1, \ldots, r_\ell \rangle \subset \hat{W}$ the Weyl group of $\mathfrak{g}$), then $\hat{W} \simeq W \times t_{Q^\vee}$ (where it should be semidirect product instead of $\times \ldots$) where $Q^\vee$ is the coroot lattice of $\mathfrak{g}$.*

Remark 5.3. For general Kac-Moody algebras, the Weyl groups are much larger, hyperbolic reflection groups.

In the affine case, $\hat{W}$ fixes level k, and $|\hat{\lambda}|$. One gets, in the intersection, paraboloids [Picture of section of hyperboloid that is a parabola].

6. Weyl character formula

Highest weight representations of Kac-Moody algebras. Let $\lambda \in \mathfrak{h}^*$, where $\mathfrak{g}(A) = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ is a Kac-Moody algebra. We define a *Verma module*

$$M(\Lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{h}+\mathfrak{n}_+)} \mathbb{C}v_\Lambda$$

where $\mathfrak{h} + \mathfrak{n}_+$ acts on V_Λ by:

$$\begin{aligned} Xv_\Lambda &= 0, && \forall x \in \mathfrak{n}_+, \\ Hv_\Lambda &= \Lambda(H)v_\Lambda, && \forall H \in \mathfrak{h} \end{aligned}$$

So $\mathbb{C}v_\Lambda$ is a $U(\mathfrak{h}+\mathfrak{n}_+)$-module,

$$\begin{array}{c} U(\mathfrak{h}+\mathfrak{n}_+) \\ \downarrow \\ U(\mathfrak{g}) \end{array}$$

By the PBW theorem, $M(\Lambda)$ has a linear $\mathbb{C}$-basis.

Let $\{F_{\alpha,i} : i = 1, \dots, \dim \mathfrak{g}_\alpha\}$ be a basis of $\mathfrak{g}_{-\alpha}$, $\forall \alpha \in \Delta_+$. Also choose a total order on Δ_+. (Some sort of lexicographical order that takes longer to write than to say.)

$$F_{\alpha_1,i_1}, F_{\alpha_2,i_2}, \dots, F_{\alpha_s,i_s}, v_\Lambda$$
$$\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_2 \text{ and if } \alpha_p = \alpha_{p+1}, i_p \leq i_{p+1}$$

We have $M(\Lambda)_\lambda = \{m | Hm = \lambda(H)m\}$ weight spaces.

$$M(\Lambda) = \bigoplus_{\lambda \in \mathfrak{h}^*} M(\Lambda)_\lambda$$

The vector v_Λ is in $M(\Lambda)_\Lambda$ by definition,

$$\begin{aligned} F_{\alpha,i} V_\Lambda &\in M(\Lambda)_{\Lambda-\alpha} \\ H(Fv_\Lambda) &= \underbrace{[H,F]v_\Lambda}_{=-\alpha(H)Fv_\Lambda} + \underbrace{FHv_\Lambda}_{=\Lambda(H)FV_\Lambda} \end{aligned}$$

So $\chi_{M(\Lambda)} = \sum_{\lambda \in \mathfrak{h}^*} \dim M(\Lambda)_\lambda e^\lambda$ is computed by counting monomials y with fixed $\sum_i \alpha_i$.

$$\chi_{M(\Lambda)} = e^\Lambda \prod_{\alpha \in \Delta_+} \frac{1}{(1-e^{-\alpha})^{\dim \mathfrak{g}_\alpha}}. \tag{6.0.1}$$

The product on Eq. 6.0.1 is called *Weyl denominator*.

Exercise 6.1. Convince yourself of this.

Example 6.2. $\mathfrak{g} = \mathfrak{sl}_2$, [Picture]

$$\begin{aligned} \chi_{M(\Lambda)} &= e^\Lambda + e^{\Lambda-\alpha} + e^{\Lambda-2\alpha} + \dots \\ &= e^\Lambda(1 + e^{-\alpha} + e^{-2\alpha} + \dots \\ &= e^\Lambda \frac{1}{1-e^{-\alpha}}. \end{aligned}$$

For certain Λ, $M(\Lambda)$ is *reducible* (i.e. there exists a submodule $0 \neq N \subset M(\Lambda)$ (with proper contention).

Lemma 6.3. *For any submodule* N*,*

$$N = \bigoplus_{\mu \in \mathfrak{h}^*} N \cap M(\Lambda)_\mu.$$

Corollary. The sum of all <u>proper</u> submodules of $M(\Lambda)$ is proper, in particular there is a maximal proper submodule.

Notation. $L(\Lambda) = M(\Lambda) \Big/ \binom{\text{max. proper}}{\text{submodule}}$

Example 6.4. $\mathfrak{sl}_2$. $\Lambda = 3\omega$ (ω: fundamental weight, $\alpha = 2\omega$.) $L(3\omega) = \mathbb{C}\left\langle e^{3\omega}, e^{-\omega}, e^{-3\omega}\right\rangle$.
[Picture]

Definition 6.5. A $\mathfrak{g}$-module is *integrable* if

- $V = \bigoplus_{\mu\in\mathfrak{h}^*} V_\mu$ (weight module).
- For all simple roots α_i; e_i and f_i are locally nilpotent on V (i.e. for all $v \in V$ there exists N such that $e_i^N v = f_i^N v = 0$.)

Remark 6.6.

- Vermas are not integrable.
- $\dim V < \infty \implies V$ integrable.
- $\mathfrak{g}$ itself (Kac-Moody) is integrable.

Dominant integrable weights. Let $\{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h}$ be the simple coroots.

Definition 6.7. The *dominant integral weights* are the weights that pair with the coroots to give integers:

$$P_+ = \{\lambda \in \mathfrak{h}^* : \langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0,1}, i = 1, \dots, \ell\}.$$

For $L(\Lambda)$ to be integrable, it is necessary that $\Lambda \in P_+$.
Indeed, suppose $L(\Lambda)$ is integrable. Then $f_i^N v_\Lambda = 0$ in $L(\Lambda)$, or rather

$$\underbrace{e_i f_i^{N+1} v_\Lambda}_{K f_i^N = 0} \in M(\Lambda),$$

and K can only be zero if $\langle \Lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}$. Applying for all i, we find $\Lambda \in P_+$ is necessary.

Proposition 6.8. *$L(\Lambda)$ is integrable if and only if $\Lambda \in P_+$.*

Proof. For the converse, use induction and Serre relations (we know the result for the highest weight, and want to prove for others). ☐

Example 6.9. $\mathfrak{sl}_3$. [Picture]

Example 6.10. $\widehat{\mathfrak{sl}_2}$. [Picture, P_+ looks like diagonal lines.]

$$\alpha_0^\vee = K - H, \qquad \alpha_1^\vee = H \in \mathfrak{sl}_2, \qquad \langle \delta, \alpha_i^\vee \rangle = 0, \quad i = 0, 1$$

Remark 6.11. For affine Kac-Moody algebras, almost nothing about the structure of $M(\Lambda)$ depends on the coefficient of δ in Λ. So it's common to consider

$$M(\Lambda) = M_k(\lambda) = M(k\Lambda_0 + \lambda), \qquad \lambda \in \mathfrak{h}^*$$

where k, the level of Λ, is super important.

Then

$$\underbrace{\hat{P}_+}_{\substack{\delta\text{-coef.}\\ =0}} = \bigcup_{k\in\mathbb{Z}_{\geq 0}} \{k\Lambda_0 + \lambda | \lambda \in P_+^k\}\{k\Lambda_0 + \lambda | \lambda \in P_+^k\}.$$

$$P_+^k = \{\lambda \in P_+ | \, \langle \lambda, \theta \rangle \leq k\} \subset P_+ \text{ for } \mathfrak{g}.$$

Consider $V = \bigoplus_{\mu\in\mathfrak{h}^*} V_\mu$ integrable, and $V_\lambda \neq 0$. Let $i \in \{1, \dots, \ell\}$. Consider $U = \bigoplus_{n\in\mathbb{Z}} V_{\lambda + n\alpha_i} \subset V$, and the action of e_i, f_i and $h_i = [e_i, f_i]$.

$\mathfrak{sl}_2 \curvearrowright U$, locally integrable. By structure of $\mathfrak{sl}_2$-represenations, U must be finite-dimensional with "symmetrical" weight space multiplicities, i.e.,

$$\{\lambda + n\alpha_i | n \in \mathbb{Z}\} \cap \{\text{weights of } V\} = \{\lambda + n\alpha_i | -p \leq n \leq q\}.$$

and

$$\langle \lambda - p\alpha_i, h_i \rangle = - \langle \lambda + q\alpha_i, h_i \rangle .$$

Consequently, the reflection $r_i(\lambda) := \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i$ has the same multiplicity as λ.

[Picture]

Now consider $M(\lambda)$ and $L(\Lambda)$ for $\Lambda \in P_+$. Actually, for general Λ, $M(\Lambda)$, while not necessarilly irreducible, has an (Ω-)composition series* by irreducibles $L(\lambda)$.

$$M(\Lambda) = V_0 \supset V_1 \supset V_2 \supset \ldots \supset V_n = 0,$$

such that V_i/V_{i+1} is $\simeq L(\lambda_i)$ (i.e. an irreducible highest weight module) for some $\lambda_i = \Lambda - \beta_i$, $\beta_i \in Q$. For Kac-Moody algebras we also consider the case that $(V_i/V_{i+1}) = 0$ for all $\mu \in \Omega + Q_+$, (which is the Ω-composition series).

[Picture.]

For an Ω-composition series, as above, we have

$$\mathrm{ch}_{M(\Lambda)} - \sum_{\lambda \geq \Omega} \underbrace{[M(\Lambda) : L(\lambda)]}_{\substack{\#\text{ of times } L(\lambda) \\ \text{appears in the compos. series}}} \mathrm{ch}_{L(\lambda)} \in \langle e^\mu : \mu \not\geq \Omega \rangle$$

Sending "$\Omega \to -\infty$", the identity

$$\mathrm{ch}_{M(\Lambda)} = \sum_{\lambda \leq \Lambda} [M(\Lambda) : L(\lambda)] \mathrm{ch}_{L(\lambda)}$$

makes sense.

Notation. $b_{\Lambda,\lambda} = [M(\Lambda) : L(\lambda)]$.

Remark 6.12. Recall the partial order on weights that $\lambda \leq \Lambda$ if $\Lambda - \lambda \in Q = \sum \mathbb{Z}_+ \alpha_i$. $b_{\Lambda,\lambda} = 1$ if $\lambda = \Lambda$ and $b_{\Lambda,\lambda} = 0$ if not ($\lambda \leq \Lambda$).

If we choose a total order on $\mathfrak{h}^*$, compatible with $\leq$. Then $\{b_{\Lambda,\lambda}\}$ is a lower triangular matrix with 1 on the diagonal. We an define $\{m_{\Lambda,\lambda}\}$ the *inverse matrix*. It's again lower triangular, 1 on the diagonal, and all $m_{\Lambda,\lambda}$ are <u>integers</u> (maybe negative now). And we have

$$\mathrm{ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} \mathrm{ch}_{M(\lambda)}. \tag{6.12.1}$$

Example 6.13. $\mathfrak{sl}_2$. $M(3) \underset{L(3)}{\supset} M(-5) \underset{L(-5)}{\supset} 0$. Since $M(-5)$ is already irreducible.

[Missing. . .]

We want to discover $m_{\Lambda,\lambda}$. In general massively difficult. For $\Lambda \in P_+$, $m_{\Lambda,\lambda}$ easy. Multiply Eq. 6.12.1 by R

$$R\mathrm{ch}_{L(\Lambda)} = \sum_{\lambda \geq \Lambda} m_{\Lambda,\lambda} e^\lambda . e^\rho R \mathrm{ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} e^{\lambda+\rho}.$$

What's ρ? It's $\rho \in \mathfrak{h}^*$ chosen so that $\langle \rho, \alpha_i^\vee \rangle = 1$, and it's called the *Weyl vector*.

Remark 6.14. For $\mathfrak{g}$ finite-dimensional, $\rho = \sum_{i=1}^{\ell} \omega_i$ necessarily. (And equals $\frac{1}{2}\sum_{\alpha \in \Delta_+} \alpha$.

For $\mathfrak{g}$ finite dimensional and the affine $\hat{\mathfrak{g}}$, $\hat{\rho} = h^\vee \Lambda_0 + \rho$ works.

For $w \in W = \langle r_i | i = 1, \ldots, \ell \rangle$, define $w(\mathrm{ch}_V) = \sum \dim V_\mu e^{W(\mu)}$. We saw $w(\mathrm{ch}_V) = \mathrm{ch}_V$ if V integrable. In particular $w(\mathrm{ch}_{L(\Lambda)} = \mathrm{ch}_{L(\Lambda)}$, $\Lambda \in P_+$.

Lemma 6.15. $m_{\Lambda,\lambda} = 0$ *unless* $\lambda + \rho = w(\Lambda + \rho)$ *for some* $w \in W$

Claim. $r_i(e^\rho R) = -e^\rho R$. So $w(e^\rho R) = \det(w) e^\rho R$ for all $w \in W$.

Proof.

$$R = \prod_{\alpha \in \Delta_+} (1 - e^{-\alpha})^{\mathrm{mult}(\alpha)}.$$

Note that

(1) $\mathrm{mult}(r_i(\alpha)) = \mathrm{mult}(\alpha)$ for all $\alpha \in \Delta$. (Since $\mathfrak{g}$ is integrable!)
(2) $r_i(\Delta_+) = \{-\alpha_i\} \cup (\Delta_+ \setminus \{\alpha_i\}$.

Any $\alpha \in \Delta_+$ is of the form $\alpha = \sum_{i=1}^{\ell} k_i \alpha_i$. If $\alpha \neq \alpha_i$, some $k_{j_0} \neq 0$, $j_0 \neq i$ and

$$\begin{aligned} r_i(\alpha) &= \sum_j k_j \alpha_j \left\langle \alpha, \alpha_i^\vee \right\rangle \alpha_i \\ &= \sum k'_j \alpha_j \\ &= e^\rho (e^{-\alpha_i} - 1) \left(\prod_{\alpha \in \Delta_+ \setminus \alpha_i} \right) \\ &= -e^\rho R. \end{aligned}$$

for $k'_{j_0} = k_{j_0} > 0$. Can't have a mixture of signs, so $r_i(\alpha) \in \Delta_\perp$.
So

$$r_i(e^\rho R) = \prod_{\alpha \in \Delta_+ \setminus \alpha_i} (1 - e^{-\alpha})^{\mathrm{mult}(\alpha)} \cdot (1 - e^{t\alpha_i} \cdot e^{\rho - \alpha_i}$$

$$r_i = \rho - \left\langle \alpha_i^\vee, \rho \right\rangle \alpha_i = \rho - \alpha_i$$

Hence

$$\sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} e^{\lambda + \rho}$$

is W-skew-invariant. (Lemma 6.15.)
Hence

$$\sum_{\lambda \leq \Lambda} m_{\Lambda,\lambda} e^{\lambda + \rho} = \sum_{w \in W} \det(w) \cdot e^{w(\Lambda + \rho)}$$

In conclusion, the *Weyl character formula*

$$\mathrm{ch}_{L(\Lambda)} = \sum_{w \in W} \det(w) \cdot \frac{e^{w(\Lambda+\rho)-\rho}}{R}$$

$\square$

Corollary (Weyl denominator formula).

$$e^\rho R = \sum_{w \in W} \det(w) e^{w(\rho)}.$$

Next time: Affine case, θ-functions. Modular forms (Poisson summation.)

7. Characters of integrable highest weight for affine Kac-Moody algebras

Can be calculated using the Weyl character formula.
Recall: $\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d$.
Dual Cartan $\hat{\mathfrak{h}}^* = \mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta$.
We consider weights $\Lambda = k\Lambda_0 + \lambda$, $\lambda \in \mathfrak{h}^*$, $k \in \mathbb{Z}_+$, $\lambda \in P_+^k$.
Simple roots: $\alpha_0 = \delta - \theta$, $\{\alpha_1, \ldots, \alpha_\ell\} \subset \mathfrak{h}^*$.
The affine Weyl group is (the *semidirect* product!)

$$\widehat{W} = \langle r_0, \ldots, r_\ell \rangle \cong W \times T$$

where $T = \{t_\alpha | \alpha \in Q\}$, where t_α is the translation

$$t_\alpha(\Lambda) = \Lambda + k\alpha - \left((\lambda, a) + k\frac{|\alpha|^2}{2}\right)\delta$$

$\Lambda = k\Lambda_0 + \lambda$.

Let's compute $\chi_{L(\Lambda_0)}$ using the Weyl character formula

$$\chi_{L(\Lambda)} = \frac{\sum_{w \in \tilde{W}} \varepsilon(w) e^{W(\Lambda+\rho)-\rho}}{R}$$

Firstly $R = \prod_{\alpha \in \hat{\Delta}} +(1 - e^{-\alpha})^{\mathrm{mult}(\alpha)}$
Recall

$$\begin{aligned} \hat{\Delta}_+ &= \{m\delta | m \in \mathbb{Z}_{\geq 1}\} \\ &\cup \{\alpha_1 + m\delta | m \in \mathbb{Z}_{\geq 0}\} \\ &\cup \{-\alpha_1 + m\delta | m \in \mathbb{Z}_{\geq 1}\} \end{aligned}$$

So let's write $e^{m\delta} = q^m$ (i.e. $q = e^{-\delta}$, is a symbol) and $y = e^{\alpha_1}$.
So

$$R = \prod_{n=1}^{\infty} \underbrace{(1 - y^{-1}q^{n-1})}_{\alpha+(n-1)\delta} \underbrace{(1 - q^n)}_{n\delta} \underbrace{(1 - yq^n)}_{-\alpha+n\delta}$$

Now let's express numerator also in terms of q and y

$$\hat{\rho} = h^\vee \Lambda_0 + \rho$$

$h^\vee$ dual Coxeter number for $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$. (For $\mathfrak{g} = \mathfrak{sl}_n$, $h^\vee = n$ and $\mathfrak{g} = E_8$, $h^\vee = 30$.)
For $\mathfrak{sl}_2$, $\rho = \frac{1}{2}\alpha_1 = \omega_1$.
$\Lambda = \Lambda_0$, $\Lambda + \rho = 3\Lambda_0 + \omega_1$.
Using the formula, we find

$$\begin{gathered} t_{m\alpha_1}(3\Lambda_0 + \omega_1) - (\Lambda_0 + \omega_1) = \\ \ldots, \Lambda_0 - 3\alpha, -2\delta, \Lambda_0, \Lambda_0 + 3\alpha_1 - 4\delta, \ldots \end{gathered}$$

[Picture]

Notation. $w(\Lambda + \rho) := w \circ \Lambda$.

One finds

$$\sum_{w\in\widehat{W}} \varepsilon(w)e^{w\circ\Lambda_0} = e^{\Lambda_0}\left(\underbrace{1+y^3q^4+y^{-3}q^2+\dots}_{+\text{ signs because } w\in T}\underbrace{-y^{-1}-y^2q^2-\dots}_{-\text{ signs because } w\in T_\sigma}\right)$$

We find explicitly

$$\chi_{L(\Lambda_0)} = e^{\Lambda_0}\frac{\sum_{m\in\mathbb{Z}} y^{3m}q^{3m^2+m} - \sum_{m\in\mathbb{Z}} y^{3m-1}q^{3m^2-m}}{\prod_{n=1}^{\infty}(1-y^{-1}q^{n-1}(1-q^n)(1-yq^n)}$$

Exercise 7.1. Put this formula in mathematica and confirm that [Picture]

$$\chi_{L(\Lambda_0)} = e^{\Lambda_0}\left(1+q(y^{-1}+1+y)+q^2(y^{-1}+2+y)+(q^3(2y^{-1}+3+2y)+\dots\right)$$

Appears that the central column here are *partitions*, i.e. $p(n) = \#$ of partitions of n. To see why recall the generating function $\sum_{n=0}^{\infty} q^n p(n) = \prod_{k=1}^{\infty}\frac{1}{1-q^k}$, and expand.

It would appear that

$$\chi_{L(\Lambda_0)} = \left(\prod_{k=1}^{\infty}\frac{1}{1-q^k}\cdot\sum_{m\in\mathbb{Z}} y^m q^{m^2}\right)$$

This identity is true, and we will see it has a vertex-algebra interpretation.

Remark 7.2. If we compute $L(0) = 1$ using the formula, we obtain

$$\begin{aligned}\prod_{n=1}^{\infty} &= (1-yq^{n-1})(1-q^n)(1-y^{-1}q^n)\\ &= \sum(\text{exercise})\end{aligned}$$

This identity is called the *Jacobi triple product identity.*

These functions

$$\sum_{n\in\mathbb{Z}} q^{n^2}, \sum_{n\in\mathbb{Z}} y^n q^{n^2}, \sum_{n\in\mathbb{Z}} y^{3n}q^{3n^2+n},\ \text{etc.}\dots$$

are all examples of *θ-functions.*

Remark 7.3. In the formula for $\chi_{L(\Lambda_0)}(y,q)$, we could put $y=1$ to get

$$\begin{aligned}\chi_{L(\Lambda_0)}(q) &= 1+3q+4q^2+7q^3+\dots\\ &= \prod\frac{1}{1-q^k}\cdot\sum_{m\in\mathbb{Z}} q^{m^2}\end{aligned}$$

If one looks at $L(\Lambda_0+\omega_1)$ (the other $\Lambda\in P_+^1$),

$$\chi_{L(\Lambda+\omega_1)} = \prod\frac{1}{1-q^k}\sum_{m\in\mathbb{Z}} q^{(m+1/2)^2-1/4}$$

8. θ-FUNCTIONS

Let's consider

$$\theta(\tau) = \sum_{n\in\mathbb{Z}} q^{n^2/2}, \qquad q = e^{2\pi i\tau}$$

This converges (absolutely on compact regions in) the comain $\mathrm{Im}(\tau) > 0$.

Consider the Fourier transform

$$\hat{g}(y) = \int_{-\infty}^{\infty} g(x) e^{2\pi i x y} dx$$

Theorem 8.1 (Poisson summation)**.**

$$\sum_{n\in\mathbb{Z}} g(n) = \sum_{n\in\mathbb{Z}} \hat{g}(n)$$

Let's take $g(x,t) = e^{-\pi t x^2}$. Then $\theta(it) = \sum_{n\in\mathbb{Z}} g(n,t)$.

In this case

$$\hat{g}(y) = \sqrt{t} e^{-\pi y^2/t}$$

(integral of Gaussian).

So we conclude that

$$\theta\left(-\frac{1}{\tau}\right) = \sqrt{\frac{\tau}{i}}\theta(\tau)$$

Note also that

$$\theta(\tau+2) = \theta(\tau)$$

because $q^{\frac{1}{2}} = e^{\pi i \tau}$.

So $\theta(\tau)$ is an example of a modular for.

What is a modular form? Let

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} | a,b,c,d \in \mathbb{Z}, \det = 1 \right\},$$

which acts on $\mathbb{H} = \{\tau \in \mathbb{C} | \mathrm{Im}(\tau) > 0\}$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}.$$

Definition 8.2. Let $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ or some subgroup. A *(weak) modular form (of weight k)* is $f : \mathbb{H} \to \mathbb{C}$ holomorphic such that

$$f\left(\frac{a\tau+b}{c\tau+d}\right) = (c\tau+d)^k f(\tau). \tag{8.2.1}$$

If we demand that Γ contains $\begin{pmatrix} 1 & N \\ 0 & 1 \end{pmatrix}$ for some $N \geq 1$, (so $f(\tau+N) = f(\tau)$, and so $f(\tau) = \sum_{n=-\infty}^{\infty} a(n) q^{2\pi i n/2}$), for $f : \mathbb{H} \to \mathbb{C}$ satisfying Eq. 8.2.1 and such that $f(\tau)$ is "meromorphic at cusps", i.e. $f(\tau) = \sum_{n \geq N_0} a(n) q^{n/N}$, etc.

Finally we would add a factor

$$f\left(\frac{a\tau+b}{c\tau+d}\right) = \varepsilon(A)(c\tau+d)^k f(\tau).$$

So in particular a weight-0 modular form is a modular function.
Denote $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. So $S\tau = -\frac{1}{\tau}$, and denote $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, so $T\tau = \tau + 1$.
Then $\Gamma = \langle S, T^2 \rangle \subset \mathrm{SL}_2(\mathbb{Z})$.
We saw

$$\theta(T^2\tau) = \theta(\tau)$$
$$\theta(S\tau) = \sqrt{\frac{\tau}{i}}\theta(\tau) = i^{-1/2}(c\tau + d)^{1/2}\theta(\tau).$$

So $\theta(\tau)$ is a modular form of weight 1/2, for the group $\Gamma = \langle S, T^2 \rangle$ with multiplier system $\varepsilon : \Gamma \to \mathbb{C}^\times$ defined by

$$\varepsilon(S) = i^{-1/2}$$
$$\varepsilon(T^2) = 1$$

Theorem 8.3. *The Dedekind eta function*

$$\eta(\tau) = q^{1/24}\prod_{n=1}^{\infty}(1 - q^n), \qquad q = e^{2\pi i \tau}$$

is also a modular form of weight $1/2$ *for* $SL_2(\mathbb{Z}) = \langle S, T \rangle$.

$$\eta\left(-\frac{1}{\tau}\right) = \sqrt{\frac{\tau}{i}}\eta(\tau)$$
$$\eta(T\tau) = e^{2\pi i/24}\eta(\tau).$$

9. Vertex algebras

(1) Recall $\hat{\mathfrak{a}} = \mathbb{C}K \oplus \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n$ the Heisenberg Lie algebra, or *oscillator Lie algebra*,

$$[h_m, h_n] = m\delta_{m,-n}K, \qquad [K, \hat{\mathfrak{a}}] = 0.$$

Remark 9.1. "It's the simplest case of an affine Lie algebra: a 1-dimensional Lie algebra with a bilinear form". It's an example of the affine Lie algebra construction $\mathfrak{g} \rightsquigarrow \hat{\mathfrak{g}}$, where now $\mathfrak{a} = \mathbb{C}$, and

$$(\cdot,\cdot) : \mathfrak{a} \times \mathfrak{a} \longrightarrow \mathbb{C}$$
$$(1,1) \longmapsto 1$$

(2) Witt Lie algebra

$$W = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}D_n,$$
$$[D_m, D_n] = (m-n)D_{m+n}.$$

The oscillator algebra $\hat{\mathfrak{a}}$ has a representation which we have seen already:

$$H = \mathbb{C}[x_1, x_2, \ldots]$$

$$h_n \mapsto \begin{cases} n\frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0 \\ 0 & \text{if } n = 0 \end{cases}, \qquad k \mapsto \mathrm{Id}$$

A picture of H [Picture].

Here I am introducing a grading of H:

$$H = \bigoplus_{n\in\mathbb{Z}_{\geq 0}} H_n, \qquad H_n = \mathrm{span}\{x_{m_1},\dots,x_{m_s} | \sum m_s = n\}.$$

Remark 9.2. $\dim(H_n) = \#\,\{\text{integer partitions of } n\} = p(n)$ and $\sum_{n=0}^{\infty}\dim(H_n)q^n = \prod_{k=1}^{\infty}\frac{1}{1-q^k}$.

Remark 9.3 (Verma module style). H can be presented alternatively as

$$H = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}1$$

where $\mathbb{C}1$ is a 1-dimensional representation of

$$\hat{\mathfrak{a}}_+ = \bigoplus_{n\geq 0}\mathbb{C}h_n \oplus \mathbb{C}K.$$

$$h_n\cdot 1 = 0 \qquad \forall n\geq 0, \qquad K\cdot 1 = 1$$

Exercise 9.4. H is irreducible.

Notation. Instead of 1 let's write $|0\rangle$.

Remark 9.5. We can easily generalise H to

$$H^\mu = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|\mu\rangle,$$

($\mu\in\mathbb{C}$), where $\hat{\mathfrak{a}}_+$ is the same, but now

$$h_n|\mu\rangle = \begin{cases} 0 & \text{if } n>0 \\ \mu\cdot|\mu\rangle & \text{if } n=0 \end{cases} \qquad\qquad K|\mu\rangle = |\mu\rangle.$$

H^μ is again a (irreducible) $\hat{\mathfrak{a}}$-module.

Generalise even more: Let $\mathfrak{h}$ be a finite-dimensional vector space, and $(\cdot,\cdot):\mathfrak{h}\times\mathfrak{h}\to\mathbb{C}$ a symmetric bilinear form. $\hat{\mathfrak{h}} = \mathfrak{h}[t,t^{-1}]\oplus\mathbb{C}K$,

$$[at^m, bt^n] = m(a,b)\delta_{m,-n}K, \qquad [K,\hat{\mathfrak{h}}] = 0.$$

Let $\mu\in\mathfrak{h}$. Define $H^\mu = U(\hat{\mathfrak{h}})\otimes_{U(\hat{\mathfrak{h}}}\mathbb{C}|\mu\rangle$, $\hat{\mathfrak{h}}_+ = \mathfrak{h}[t]\oplus\mathbb{C}K$, and

$$at^m\cdot|\mu\rangle = \begin{cases} (\mu,a)|\mu\rangle & \text{if } m=0 \\ 0 & \text{if } m>0. \end{cases}$$

Returning to $\hat{\mathfrak{a}}\curvearrowright H = H^\circ$. Introduce

$$L_m = \frac{1}{2}\sum_{k\in\mathbb{Z}} h_{m-k}h_k.$$

If $m\neq 0$, this sum is well-defined in $\mathrm{End}(H)$.

Indeed, h_{m-k} and h_k commute ($m\neq 0$), and $\forall p(\underline{x})\in H$, there exists N such that $p(\underline{x}) = p(x_1,\dots,x_{N-1})$, then $h_k\cdot p(x) = 0$ $\forall k\geq N$, and $h_{m-k}\cdot p(x) = 0$ $\forall m-k\geq N$.

So $(\sum h_{m-k}h)k)p(\underline{x})$ is a finite sum. Of course, the number of terms in the sum depends on $p(x)$ and can be arbitrarily large.

Exercise 9.6. If m, n and $m+n$ are not zero, then

$$[L_m, L_n] = (m-n)L_{m+n}$$

holds in $\mathrm{End}(H)$.

(Trying to be a representation of W

$$D_n \mapsto L_n \in \mathrm{End}(H).)$$

But $\sum_{k\in\mathbb{Z}} h_{-k}h_k$ is not well-defined. Indeed,

$$\begin{aligned}\left(\sum_{k\in\mathbb{Z}} h_{-k}h_k\right)1 &= \sum_{k\geq 1} k\frac{\partial}{\partial x_k}(x_k 1)\\ &= (1+2+3+4+\dots)1\end{aligned}$$

!

Normal ordering idea ("that physicist do"). Let's cheat and redefine the product as

$$: h_{-k}h_k := \begin{cases} h_{-k}h_k & \text{if } k\geq 0\\ h_k h_{-k} & \text{if } k<0.\end{cases}$$

Now $\sum_{k\in\mathbb{Z}} : h_{-k}h_k :$ is well-defined!

Notation. Let us consider series of the form

$$a(z) = \sum_{n\in\mathbb{Z}} a_{(n)}z^{-n-1}, \qquad a_{(n)} \in \mathrm{End}(V),$$

where V is a vector space.

Definition 9.7 (Most important in the course)**.** We say $a(z)$ is a *quantum field* on V if, for every $v \in V$ there exists $N \in \mathbb{Z}$ such that $a_{(n)}v = 0\ \forall n \geq N$.

Example 9.8. $V = H = \mathbb{C}[x_1, x_2, \dots]$,

$$a(z) = h(z) = \sum_{n\in\mathbb{Z}} h_n z^{-n-1}.$$

$$h(z) = \dots + 3z^{-4}\frac{\partial}{\partial x_3} + 2z^{-3}\frac{\partial}{\partial x_1} + x_1 + x_2 z + x_3 z^2 + \dots$$

Any fixed $v \in H$ is $v = p(x_1, \dots, x_{N-1})$ for some N. Then $h_N v = N\frac{\partial}{\partial x_N}v = 0$.

So $h(z)\curvearrowright H$ is a quantum field.

Definition 9.9. For a quantum field (or any series in fact) $a(z) = \sum_{n\in\mathbb{Z}} a_{(n)}z^{-n-1}$, define the *creation* and *annihilating operators*

$$a(z)_+ = \sum_{n\leq -1} a_{(n)}z^{-n-1}, \qquad a(z)_- = \sum_{n\geq 0} a_{(n)}z^{-n-1}.$$

The quantum field condition solves the problem of the infinite series...

The normally order product of quantum fields $a(z) = \sum_{n\in\mathbb{Z}} a_{(n)}z^{-n-1}$, and $b(z) = \sum_{n\in\mathbb{Z}} b_{(n)}z^{-n-1}$ is.

$$: a(z)b(z) := a(z)_+ b(z) + b(z)a(z)_-.$$

Exercise 9.10 (Most important of the course)**.** Show that if $a(z)$ and $b(z)$ are quantum fields, then $: a(z)b(z) :$ is well-defined, and is a quantum field.

Next idea: in our example of $h(z)\curvearrowright H$, the coefficients h_n were coming from a Lie algebra, so we had relations $[h_m, h_n] = (\ldots)$. (Indeed, $[h_m, h_n] = m\delta_{m,-n}\mathrm{Id}_H$ in this case.) We sould like to interpret such relations at the level of $h(z)$.

$$[h(z), h(z)] = \sum_{m,n\in\mathbb{Z}} \qquad = \sum_{m,n} z^{-(m+n)-2}$$

$$= \sum z^{-p-2}\left(\underbrace{\sum_{m+n=2} p[h_m, h_n}_{\text{infinite?}}\right)$$

which is a bad idea, since that sum can be infinite. Better idea: change a variable:

$$[h(z), h(w)] = \sum_{m,n\in\mathbb{Z}} [h_m, h_n] z^{-m-1} w^{-n-1}.$$

In this example,

$$\begin{aligned}[h(z), h(w) &= \sum_{m,n\in\mathbb{Z}} m\delta_{m,-n} z^{-m-1} w^{-n-1} I_H \\ &= \sum_{m\in\mathbb{Z}} z^{-m-1} w^{m-1} I_H.\end{aligned}$$

Observe (geometric series):

$$\begin{aligned}\frac{1}{z-w} &= \sum_{k\geq 0} z^{-k-1} w^k \qquad \text{(convergent for } |z| > |w|) \\ \frac{1}{z-w} &= -\sum_{k\geq 0} w^{-k-1} z^k \\ &= -\sum_{k<0} z^{-k-1} w^k \qquad \text{(convergent for } |z| < |w|)\end{aligned}$$

So, in a sense

$$\sum_{k\in\mathbb{Z}} z^{-k-1} w^k \text{``}=\text{''} \frac{1}{z-w} - \frac{1}{z-w}.$$

This motivates us to introduce the expression

Definition 9.11. The *formal delta function* is

$$\delta(z, w) = \sum_{k\in\mathbb{Z}} z^{-k-1} w^k \in \mathbb{C}[\![z, z^{-1}, w, w^{-1}]\!].$$

Why (Dirac) delta function? Recall that $\delta(x)$ is the *distribution* on $\mathbb{R}$ defined by $\delta[f(x)] = f(0)$ for all test functions $f(x)$. Every function $k(x)$ gives a distribution D_k.

$$D_k[f] = \int_{-\infty}^{\infty} k(x) f(x) dx,$$

$f \in C_c^\infty(\mathbb{R})$.

If δ were of the form D_k (it's not) it would have to look like [Picture, positive part of y axis, all x axis.

One can show that (it's a theorem by Plamelj)

$$\delta(x) = \frac{1}{2\pi i} \lim_{\varepsilon \to 0_+} \left(\frac{1}{x - i\varepsilon} - \frac{1}{x - i\varepsilon} \right)$$

as distributions (i.e. limit taken in "distributional sense".)

(So that's why we called the delta function that way.)

Denote by $i_{z,w}$ the "expansion in positive powers of w". E.g.

$$i_{z,w} \frac{1}{z-w} = \sum_{k \geq 0} z^{-k-1} w^k,$$

similarly,

$$i_{z,w} \frac{1}{(z-w)^2} = \sum_{k \geq 0} k z^{-k-1} w^{k-1},$$

$$i_{z,w}(w^{-2}) = w^{-2}.$$

Exercise 9.12. So, in fact,

$$[h(z), h(w)] = i_{z,w} \frac{1}{(z-w)^2} - i_{w,z} \frac{1}{(z-w)^2} \qquad\qquad = \partial_w \delta(z, w).$$

What exactly is $i_{z,w}$?

Notation.

$\mathbb{C}[z]$	ring of polynomials
$\mathbb{C}[z, z^{-1}]$	ring of Laurent polynomials
$\mathbb{C}[\![z]\!]$	ring of power series
$\mathbb{C}[\![z, z^{-1}]\!]$	vector space (not ring!) of formal distributions
$\mathbb{C}(\!(z)\!)$	field of Laurent series

where the last is $\sum_{n \geq N} f a_n z^n$ for some N.

Since $\mathbb{C}(\!(z)\!)$ is a field, $\mathbb{C}(\!(z)\!)(\!(w)\!)$ is also a field.

There are natural inclusions

$$\mathbb{C}[z, w] \to \mathbb{C}(\!(z)\!)(\!(w)\!) \to \mathbb{C}[\![z^{\pm 1}, w^{\pm 1}]\!].$$

Let $\mathbb{C}(z, w)$ denote the fraction field of the domain $\mathbb{C}[z, w]$. By the property of $\mathbb{C}(z, w)$, there exists an embedding

$$\begin{tikzcd} \mathbb{C}[z,w] \arrow[rr] \arrow[dr] & & \mathbb{C}(\!(z)\!)(\!(w)\!) \\ & \mathbb{C}(z,w) \arrow[ur, "i_{z,w}"'] & \end{tikzcd}$$

The following diagram does not commute: [diagram]

We computed

$$[h(z), h(w)] = \partial_w \delta(z, w)$$

We can similarly compute

$$h(z)h(w) =: h(z)h(w) : + i_{z,w} \frac{1}{(z-w)^2}$$

$$h(w)h(z) =: h(z)h(w) : + i_{w,z} \frac{1}{(z-w)^2}.$$

Notation. We write $\partial_w = \frac{\partial}{\partial w}$ and $\partial_w^{(j)} = \frac{1}{j!}\partial_w^j$.

Lemma 9.13. (1) *If we multiply the delta function with $(z-w)$ we get zero, that is,*
$$(z-w)\delta(z,w) = 0.$$
(2)
$$i_{z,w}\frac{1}{(z-w)^{j+1}} - i_{w,z}\frac{1}{(z-w)^{j+1}} = \partial_w^{(j)}\delta(z,w).$$
(3) $\forall j \geq 0$,
$$(z-w)^{j+1}\partial_w^{(j)}\delta(z,w) = 0.$$
(4) *Whenever $m \leq j$,*
$$(z-w)^m\partial^{(j)}(z,w) = \partial_w^{(j-m)}\delta(z,w).$$

Proof. □

Remark 9.14. All this is completely parallel to the Dirac δ-distribution $\delta(x)$.
$$x\delta(x) = 0, \qquad x\delta'(x) = \delta(x), \text{ etc.}, x^2\delta'(x) = 0.$$

10. The residue pairing

Let $f(z) \in \mathbb{C}[z, z^{-1}]$, which is a vector space with basis $\{z^n : n \in \mathbb{Z}\}$.

An element of the dual vector space is a formal linear combination
$$\sum_{n\in\mathbb{Z}} c_n\varphi_n, \qquad \text{where } \varphi_n(z^k) = \begin{cases} 0 & \text{if } k \neq n \\ 1 & \text{if } k = n \end{cases}$$
(no restriction on the $c_n \in \mathbb{C}$).

Identify $\sum c_n\varphi_n$ with
$$c(z) = \sum c_n z^{-n-1} \in \mathbb{C}[z^{\pm 1}$$
so that φ acts on $f(z) = \sum f_n z^n \in \mathbb{C}[z^{\pm 1}]$, as
$$\varphi(f) = \sum_n c_n f_n = \mathrm{Res}_z c(z)f(z)dz,$$
where

Definition 10.1. If U is a vector space and $a(z) = \sum_n a_n z^n \in U[\![z^{\pm 1}]\!]$,
$$\mathrm{Res}_z a(z)dz = a_{-1}$$

Let's record a few properties of $\mathrm{Res}_z(\cdot)dz := \mathrm{Res}_z(\cdot)$.

Lemma 10.2. (1) *$Res_z f(z)\delta(z,w) = f(w)$.*
(2) *$Res_z f(z)(\partial_z g(z)) = -Res_(\partial_z f(z))fg(z)$.*

Proof. (1) Straightforward computation.
(2) Product rule, and uses that derivatives have no residues.

□

So, in our example, $H, h(z)$ we see that
$$(z-w)^2[h(z), h(w)] = 0,$$
which is a nontrivial example: we really need to take the square, and also the bracket is not zero.

Definition 10.3. Two quantum fields $a(z)$ and $b(z)$ on a vector space V are *mutually local* if there exists N such that

$$(z-w)^N[a(z),b(w)]=0.$$

Remark 10.4. See [**?**, Chapter 1] for physics motivation. More generally, if $D=\sum_{m,n}D_{m,n}z^m w^n\in U[\![z^{\pm 1},w^{\pm 1}]\!]$, we call D *local* if $(z-w)^N D=0$ for some N.

Proposition 10.5. *Let $a(z),b(z)$ be a local pair of quantum fields on V. Then*

$$a(z)b(w)=:a(z)b(z):+\sum_{j=0}^{N-1}c_j(w)i_{z,w}\frac{1}{(z-w)^{j+1}} \tag{10.5.1}$$

and

$$b(w)a(z)=:a(z)b(z):+\sum_{j=0}^{N-1}c_j(w)i_{w,z}\frac{1}{(z-w)^{j+1}} \tag{10.5.2}$$

In particular,

$$[a(z),b(w)]=\sum_{j=0}^{N-1}c_j(w)\partial_w^{(j)}\delta(z,w). \tag{10.5.3}$$

Furthermore

$$c_j(w)=Res_z(z-w)^j[a(z),b(w)]. \tag{10.5.4}$$

Proof. First we prove 10.5.3 using 10.5.4. □

Exercise 10.6. Deduce 10.5.1 and 10.5.2 from 10.5.3.

Notation. If $a(w)$ and $b(w)$ are quantum fields on V, we denote

$$a(w)_{(j)}b(w)=\mathrm{Res}_z(z-w)^j[a(z),b(w)]$$

for $j\in\mathbb{Z}_{\geq 0}$.

Example 10.7. On H, $[h(z),h(w)]=\partial_w\delta(z,w)I_H$. So $\mathrm{Res}_z\partial_w\delta(z,w)I_H=0$ and $\mathrm{Res}_z(z-w)\partial_w\delta(z,w)I_H=\mathrm{Res}_z\delta(z,w)I_H=I_H$.

Note that I_H is a (very simple) quantum field.

The following definition generalizes the j-product to any integer, possible negative.

Definition 10.8. For $n\in\mathbb{Z}$, and $a(w)$, $b(w)$ quantum fields, we define the n^{th} *product* $a(w)_{(n)}b(w)$ as

$$a(w)_{(n)}b(w)=\mathrm{Res}_z[i_{z,w}(z-w)^n a(z)b(w)-i_{w,z}(z-w)^n b(w)a(z)].$$

Remark 10.9.

(1) We recover the prior definition: if $n\geq 0$,

$$i_{z,w}(z-w)^n=i_{w,z}(z-w)^n=\sum_{r=0}^{n}\binom{n}{r}z^{n-r}(-w)^r,$$

a finite sum.

(2) (Exercise.) If $n=1$, then

$$\begin{aligned}a(w)_{(-1)}b(w)&=a(w)_+b(w)+b(w)a(w)_-\\&=:a(w)b(w):\end{aligned}$$

(3) (Exercise.) The more negative products are not something new: for $k \geq 0$,

$$a(w)_{(-k-1)}b(w) =:(\partial_w^{(k)}a(w))b(w):$$

We have actually already used the following proposition:

Proposition 10.10. *If $a(w), b(w)$ are quantum fields, then $a(w)_{(n)}b(w)$ is also a quantum field for all $n \in \mathbb{Z}$.*

Definition 10.11. A *vertex algebra* consists of a vector space V, a set $\mathcal{F}$ of quantum fields on V, a nonzero vector $|0\rangle \in V$, and a linear map $T : V \to V$, such that

(1) $T|0\rangle = 0$, and $[T, a(z)] = \partial_z a(z)\ \forall a(z) \in \mathcal{F}$.
(2) V is spanned by $a^{i_1}_{(n_1)}, \ldots, a^{i_s}_{(n_s)}|0\rangle$, where $a^{i_j}(z) \in \mathcal{F}$ (and $s(z) = \sum a_{(n)}z^{-n-1}$ always).
(3) All pairs $a(z), b(z) \in \mathcal{F}$ are mutually local. (We saw (ref?) that this is equivalent to $[a(z), b(w)] = \sum_{j=0}^{N-1} c_j(w)\partial_w^{(j)}\delta(z,w)$ and described the coefficients c_j, ...)

We call $|0\rangle$ the *vacuum vector* and T the *translation operator*.

In fact, our Heisenberg example that we've been discussing so far is an example:

Example 10.12. $V = H = \mathbb{C}[x_1, x_2, \ldots]$, $\mathcal{F} = \{h(z)\}$, $|0\rangle = 1$, $T =?$, is a vertex algebra.

Answer 1. Recall $L(z) = \frac{1}{2} :h(z)h(z):= \frac{1}{2}h(z)_{(-1)}h(z)$, $L(z) = \sum_{m\in\mathbb{Z}} L_m z^{-m-z}$. In particular, $L_{-1} = \frac{1}{2}\sum_{k\in Z} h_k h_{-1-k}$. Well, $T = L_{-1}$.

Answer 2. Given that T must satisfy $T1 = 0$, and $[T, h(z)] = \partial_z h(z)$, i.e.

$$[T, h_n] = -nh_{n-1},$$

and H is generated from 1 by $\{h_n | n \leq -1\}$, the action of T on H (if well-defined) is completely determined.

Exercise 10.13. Write a formula for $T(x_1^{m_1}x_2^{m_2}\ldots x_s^{m_2})$.

11. A second definition of vertex algebra

Now we aim for a second definition of vertex algebras.

Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. Define

$$\tilde{\mathcal{F}} = \{\text{quantum fields } a(z) \text{ on } V | [T, a(Z)] = \partial_z a(z)\}$$

$$\overline{\mathcal{F}} = \bigcup_{k\geq 0} \mathcal{F}_k, \qquad \mathcal{F}_0 = \{I_V\}, \qquad \mathcal{F}_k = \{a(z)_{(n)}b(z) | a(z) \in \mathcal{F}, b(z), \mathcal{F}_{k-1}\}$$

$$\mathcal{F}' = \{b(z) \in \tilde{\mathcal{F}} | a(z), b(z) \text{ is local}, \forall a(z) \in \mathcal{F}\}$$

Where I_V is the identity of V considered as a quantum field. The idea is that this is like taking a subalgebra and taking the commutator over and over again.

Lemma 11.1. $a(z)_{(-1)}I_V = a(z)$.

Proof. Direct calculation. □

Lemma 11.2. $a(z)_{(-n-1)}I_V = \partial_z^{(n)}a(z)$.

Proof. Similar. □

Lemma 11.3 (Dong)**.** *Suppose $a(z), b(z)$ and $c(z)$ are mutually local in pairs. Let $n \in \mathbb{Z}$. Then $a(z)_{(n)}b(z)$ and $c(z)$ is a local pair.*

Proof. Done in lecture. □

Therefore

$$\mathcal{F} \subset \overline{\mathcal{F}} \overset{\text{Dong}}{\subset} \mathcal{F}' \subset \tilde{\mathcal{F}}$$

Remark 11.4. To be sure, we should chec $\overline{\mathcal{F}} \subset \tilde{\mathcal{F}}$.

F tilde is the translation invariant

Exercise 11.5. If $[T, a(z)] = \partial_z a(z)$, $[T, b(z)] = \partial_z b(z)$, then

$$[T, a(z)_{(n)} b(z)] = \partial_z (a(z)_{(n)} b(z)).$$

Theorem 11.6. (1) *For any* $a(z) \in \tilde{\mathcal{F}}$,

$$\begin{aligned} a(z)|0\rangle &= v + z \cdot Tv + \frac{z^2}{2} T^2 v + \dots \\ &= e^{2T} \qquad (v \in V) \end{aligned}$$

Thus we define

$$\begin{aligned} s : \tilde{\mathcal{F}} &\longrightarrow V \\ a(z) &\longmapsto a(z)|0\rangle|_{z=0} \end{aligned}$$

(2)

$$s(z(z)_{(n)} b(z) = a_{(n)} s(b(z)).$$

(3) *Let* $\mathcal{G} \subset \tilde{\mathcal{F}}$ *be such that* $s(\mathcal{G}) = V$, *and suppose* $a(z)$ *is local with all* $b(z) \in \mathcal{G}$. *"If you commute with a large enough bunch of guys, you are close enough to being zero." If* $s(a(z)) = 0$, *then* $a(z) = 0$.

Proof. (1) We need to prove $a(z)|0\rangle$ does not have negative powers of z. So suppose it does, i.e. there is $n \geq 0$ such that $a_{(n)}|0\rangle z^{-n-1} \neq 0$.

Idea: use translation invariance to pair n and $-n$, and then the fact that $a(z)$ is a quantum field to get some vanishing.

So first we translate and find that:

$$Tv(z) = Ta((z)|0\rangle = (Ta(z) - a(z)T)|0\rangle = \partial_z a(z)|0\rangle$$

so that

$$v_n = \frac{1}{n} T v_{n-1}, \qquad (n \neq 0). \tag{11.6.1}$$

So, $v_n \neq 0$ for some $n < 0 \implies v_{n-1}$ also not zero. But we also know by $a(z)$ being a quantum field that there exists N such that $a_{(n)}|0\rangle = 0$ for all $n \geq N$. So, we'd have a contradiction. So $v(z) = \sum_{n \geq 0} v_n z^n$.

Now Eq. 11.6.1 also implies

$$v_1 = Tv_0, v_0 = \frac{1}{2} T v_1, \dots, v_n = \frac{1}{n!} T^n v_0.$$

So $a(z)|0\rangle = e^{zT} v_0$ $(v_0 \in V)$.

(2) Consider

$$a(w)_{(n)} b(w)|0\rangle = \mathrm{Res}_z (a(z) b(w) i_{z,w} (z-w)^n - \underbrace{b(w) a(z) i_{w,z} (z-w)^n}_{=0}$$

where that vanishing is because the $i_{w,z}$ expands the argument in positive powers, and $b(w)a(z)$ also has only positive powers since $a(z)|0\rangle$ also has only positive powers (and the residue picks the coefficient of the power -1). So we obtain

$$= \mathrm{Res}_z a(z)b(w)i_{z,w}(z-w)^n.$$

But

$$\begin{aligned} s(a(w)_{(n)}b(w)) &= (\mathrm{Res}_z s(z)b(w)i_{z,w}(z-w)^n|0\rangle)\,|_{w=0} \\ &= \mathrm{Res}_z a(z)b(w)z^n|0\rangle|_{z=0} \\ &= \mathrm{Res}_z z^n a(z)(s(b(w))) \\ &= \mathrm{Res}_z z^n \sum_k a_{(k)}z^{-k-1}s(b) \\ &= a_{(n)}s(b). \end{aligned}$$

(3) By locality, there exists N such that

$$\begin{aligned} (z-w)^N a(z)b(w)|0\rangle &= (z-w)^N b(w)a(z)|0\rangle \\ &= (z-w)^N b(w)e^{2T}s(a) = 0 \end{aligned}$$

$$\begin{aligned} (z-w)^N a(z)b(w)|0\rangle|_{w=0} &= 0 \\ \implies z^N a(z)s(b) &= 0 \\ \implies a(z)v &= 0 \\ \implies a(z) &= 0 \end{aligned}$$

$\square$

Corollary.

(1) $S|_{\mathcal{F}'}$ is injective.
(2) $S|_{\overline{\mathcal{F}}}$ is surjective.
(3) $S|_{\mathcal{F}}$ is isomorphism $\overline{\mathcal{F}} \to V$.
So we define the inverse (linear) map $Y : V \to \overline{\mathcal{F}}$, i.e. for all $a \in V$, $Y(a,z)$ denotes $s^{-1}(a) \in \overline{\mathcal{F}}$.
(4) $(V, |0\rangle, T, \overline{\mathcal{F}})$ is a vertex algebra, and

$$Y(a_{(n)}b, z) = Y(a,z)_{(n)}Y(b,z) \qquad \forall a, b \in V.$$

So we have

$$\begin{tikzcd} \mathcal{F} \arrow[r] & \overline{\mathcal{F}} \arrow[r] \arrow[d, two heads] & \mathcal{F}' \arrow[r] \arrow[d, hook] & \tilde{\mathcal{F}} \arrow[d, "s"] \\ & V \arrow[r, "\mathrm{id}"] & V \arrow[r, "\mathrm{id}"] & V \end{tikzcd}$$

Proof. (1) Now we apply part 3 in the theorem to $a(z) \in \mathcal{F}'$, $\mathcal{G} = \overline{\mathcal{F}}$. If $s(a(z)) = 0$, then $a(z) = 0$. So $s : \mathcal{F}' \to V$ is injective (as in the diagram).

(2) By iterating the property in the theorem that $a_{(n)}s(b(z)) = s(a(z)_{(n)}b(z))$ so we can take out "one by one" in the following product to obtain

$$a^1_{(n_1)} \dots a^s_{(n_s)}|0\rangle = s(a^1(z)_{(n)} \dots a^s(z)_{(n_s)}I_V$$

This proves the double headed arrow in the diagram.

(3) Now define $Y(-,z) : V \to \overline{\mathcal{F}}$.

$$s(Y(a,z)_{(n)}Y(b,z)) = a_{(n)}s(Y(b,z)) = a_{(n)}b.$$

(4) Then all the axioms in our definition of vertex algebra are satisfied!

□

Remark 11.7. Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. $\overline{\mathcal{F}}$ and Y as above. $\forall a, b \in V$,

$$[Y(a,z), Y(b,w)] = \sum_{j=0}^{N-1} c_j(w)\partial_w^{(j)}\delta(z,w). \tag{11.7.1}$$

$c_j(w) = Y(a,w)_{(j)}Y(b,w) = Y(a_{(j)}b, w)$.

Which is like something we had written before. But now we know more. Let us extract the z^{-m-1}, w^{-n-1} coefficient of Eq. 11.7.1. We obtain

$$LHS = \sum_{m,n}[a_{(m)}z^{-m-1}, b_{(n)}w^{-n-1}] = [a_{(m)}, b_{(n)}]$$

$$RHS = \sum_j \left(\sum_k (a_{(j)}b)_{(k)}w^{-k-1}\right)\left(\sum_s \binom{s}{j} z^{-s-1}w^{s-j}\right)$$

The coefficient of $z^{-m-1}w^{-n-1}$ of RHS is, $s = m$,

$$\sum_j \binom{m}{j} w^{m-j}(a_{(j)}b)_{(k)}w^{-k-1} = \sum_j \binom{m}{j}(a_{(j)}b)_{(m+n-j)}$$

So we put this as a proposition:

Proposition 11.8 (Commutator formula)**.** *In a vertex algebra* $(V, |0\rangle, T, Y)$*, we have*

$$[a_{(m)}, b_{(n)}] = \sum_{j\geq 0}\binom{m}{j}(a_{(j)}b)_{(m+n-j)} \qquad \forall a, b \in V, \quad m, n \in \mathbb{Z}$$

Exercise 11.9. Prove that in a vertex algebra

$$[Y(a,z)Y(b,w)i_{z,w} - Y(b,w)Y(a,z)i_{w,z}](z-w)^n = \sum_{j\geq 0} Y(a_{(n+j)}b, w)\partial_w^{(j)}\delta(z,w). \tag{11.9.1}$$

Hint. Prove that he left hand side is local.

By extracting coefficients of Eq. 11.9.1, we obtain for all $a, b, c \in V$ and $m, n, k \in \mathbb{Z}$:

$$\begin{aligned} &\sum_{j\geq 0}\binom{m}{j}(a_{(n+j)}b)_{(m+k-j)}c \\ &= \sum_{j\geq 0}(-1)^j\binom{n}{j}\left(a_{(m+n-j)}b_{(k+j)}c - (-1)^n b_{(n+k-j)}a_{(n+j)}c\right) \end{aligned} \tag{11.9.2}$$

This is called *Bocherd's identity*, and is the key ingredient in our second definition of vertex algebra:

Definition 11.10. A *vertex algebra* is a vector space V, a nonzero vector $|0\rangle \in V$, and a set of bilinear products $V \times V \to V$, $a, b \mapsto a_{(n)}b$, $n \in \mathbb{Z}$ such that

(1) $\forall a, b \in V$ $\exists N$ such that $a_{(n)}b = 0$ $\forall n \geq N$.
(2) $\forall a \in V$, $|0\rangle_{(n)}a = \delta_{n,-1}a$.
(3) $\forall a \in V$, $a_{(-1)}|0\rangle = a$ and $a_{(\geq 0)}|0\rangle = 0$.
(4) Equation 11.9.2 holds $\forall a, b, c \in V$, $m, n, k \in \mathbb{Z}$.

References

[Kac90] V.G. Kac, *Infinite-dimensional lie algebras*, Progress in mathematics, Cambridge University Press, 1990.